

Reproductive Outcomes in Women with Prior Cesarean Section Undergoing *In Vitro* Fertilization: A Retrospective Case-control Study

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Summary: The impact of prior cesarean section (CS) on the pregnancy and neonatal outcomes of *in vitro* fertilization and embryo transfer (IVF-ET) was investigated. A retrospective analysis was performed on 144 patients with prior CS between January 2013 and December 2015. The pregnancy, delivery, and neonatal outcomes of patients who had previous CS delivery and received IVF-ET were analyzed. The control group comprised 166 patients who had only previous vaginal delivery (VD) and received IVF-ET during the same period. The results showed that the basal follicle stimulating hormone level, estradiol level on human chorionic gonadotropin (hCG) day, gonadotrophin dosage, duration of stimulation, retrieved oocytes, fertilization rate, high-quality embryo rate, multiple birth rate, abortion rate and ectopic pregnancy rate had no significant difference between the two groups ($P>0.05$). The pregnancy rate (40.28% vs. 54.22%) and implantation rate (24.01% vs. 34.67%) were significantly lower ($P<0.05$), and the ratio of embryo difficulty transfer (9/144 vs. 0/166) was significantly higher in CS group than in VD group. The risk of pernicious placenta previa and postpartum hemorrhage in twin deliveries was significantly increased in CS group as compared with that in VD group ($P<0.05$), and gestational age and neonatal birth weight were significantly reduced in twin deliveries as compared with singleton deliveries in both groups ($P<0.05$). It was suggested that the existence of CS scar may impact embryo implantation and clinical pregnancy outcome, and increase the difficulty of ET. We should limit the number of transfer embryos to avoid multiple pregnancies and strengthen gestational supervision in patients with cesarean scar.

Key words: cesarean section; *in vitro* fertilization and embryo transfer; pregnancy; complication

Since the past decades, there has been a continuous increase in rates of delivery through cesarean section (CS), resulting in increasing obstetrical sequelae, which has attracted much attention^[1, 2]. CS is not only associated with complications in subsequent pregnancies, such as cesarean scar pregnancy with life-threatening bleeding, placenta previa, placenta accreta, scar dehiscence or uterine rupture due to the incomplete healing of the CS incision but also with a marked increase in the risk of subfertility and abnormal uterine bleeding after CS^[3, 4].

With the abolition of the single-child policy in China, the number of secondary infertility patients with a history of CS choosing *in vitro* fertilization/intracytoplasmic sperm injection-embryo transfer (IVF/ICSI-ET) as an alternative strategy for their second progeny has markedly increased. Although there are a significant number of articles on the subsequent spontaneous pregnancy risks of CS, there is little discussion regarding the influence of CS scars on reproductive outcome of IVF/ICSI-ET. Naji *et al* reported that a CS scar can reduce the chance of embryo implantation and lead to spontaneous abortion^[5]. A deleterious effect of the

presence of a CS scar on the uterus was reported by Ben-Nagi, who suggested there might be a link between altered uterine immunobiology, CS and embryo implantation^[6]. Contrarily, Zhang *et al* reported that the presence of a CS scar did not affect pregnancy outcomes after IVF^[7]. This raises questions on what role the previous CS plays in affecting clinical pregnancy outcomes and avoiding the complications of IVF/ICSI-ET.

In the recent years, CS scar defects are increasingly described by transvaginal ultrasonography, and the reported incidence of CS scar defects is as high as 61% after one CS, reaching 100% in women undergoing at least three^[8]. Cesarean-delivery scar defects, being a sign of poor healing after CS, are currently underdiagnosed and may consequently be left untreated at a staggering rate^[9]. This defect comprises a thinning and indentation of the myometrium at the site of hysterotomy on the anterior uterine wall. On the other hand, multiple numbers of ET in IVF/ICSI possibly increase the incidence of iatrogenic multiple pregnancies as well as increase the embryo implantation rate, but excessive uterine distension of multiple pregnancy increases the risk of uterine rupture and maternal-fetal complications for patients who had a previous CS defect^[10].

Histological changes in the healing myometrium

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were described by Wojdecki *et al* who presented a case of a 31-year-old woman who had hysterectomy due to massive hemorrhage 18 days after CS^[11]. During histological evaluation of the myometrial tissue which surrounded previous uterine incision, they found necrotic muscular fibers, granuloma, fibroblasts, fibrocytes and fibrous tissue. A series of 51 hysterectomy specimens due to uterine bleeding studied by Morris in 1995 found significant pathological changes, including distortion and widening of the lower uterine segment (75%), “overhang” of congested endometrium above the scar recess (61%), polyp formation conforming to the contours of the scar recess (16%), fragmentation and breakdown of the endometrium of the scar (37%) present in the area of the post CS scar^[12]. These previous researches hint that these pathological changes may increase the difficulty of ET in IVF/ICSI-ET.

The purpose of the present study was to investigate whether the prior CS would have an impact on reproductive outcomes and ET procedure during IVF/ICSI-ET. The long-term effects of CS need to be further elucidated to provide more information to patients undergoing IVF/ICSI-ET.

1 MATERIALS AND METHODS

1.1 Patient Population

A retrospective case-control study was conducted on a total of 144 female patients with a history of single or multiple CS in our reproductive center from January 2013 to December 2015. A total of 166 patients with only previous vaginal delivery (VD) during the same period served as controls. All participants met the following criteria: (1) the first cycle of IVF/ICSI and fresh ET performed; (2) the ages of the patients ranging from 20 to 37 years and body weight of at least 40 kg; (3) administration of long mid-luteal gonadotropin releasing hormone (GnRH) agonist protocol; (4) tubal factor-induced infertility or male infertility. The exclusion criteria were as follows: (1) a history of unilateral oophorectomy, myomectomy, or resection of endometrioma; (2) congenital or acquired uterine malformations, uterine submucous fibroids, adenomyosis, intrauterine adhesion; (3) abnormal results on parental karyotyping; (4) the recurrent spontaneous abortion history (defined as two or more previous spontaneous pregnancy losses). The study was approved by the Medical Ethics Committee of Renmin Hospital of Wuhan University. When admitted to the hospital, all participants were formally informed of the potential academic use of their clinical data in the future and the written informed consents were obtained from all participants.

1.2 Stimulation Protocol and IVF Procedure

In all the cases, a long mid-luteal GnRH agonist protocol was performed for superovulation. Down-regulation was carried out using daily GnRH agonist (triptorelin, s.c. 0.1 mg, Ferring, Pharmaceuticals, Kiel, Germany) beginning on day 21 of the previous cycle, as confirmed by ultrasound and blood test. After 2–3 weeks of down-regulation, confirmed by a blood test and ultrasound, Gn (Gonal-F, i.m., 75 IU or 450 IU,

Merck-Serono, Aubonne, Switzerland) was administered at 112.5–300 IU/day starting on cycle days 5–8 of stimulation. Gn dose was adjusted according to ovarian response. All patients were monitored using transvaginal ultrasound and serum ovarian steroid hormone concentrations during superovulation. Final oocyte maturation was triggered when at least three dominant follicles ≥ 17 mm were present on ultrasound, with the administration of 6000–10 000 IU human chorionic gonadotropin (hCG) injection (hCG, 1000 IU, Livzon Pharmaceuticals Co., China). Transvaginal oocyte aspiration was performed 36 h later by ultrasound-guided follicle puncture. The fertilization technique (short-term fertilization or ICSI) was selected according to the kinetic analysis of the washed sperm suspension. No more than 2 embryos were selected for ultrasound-guided transfer on day 3 or day 5 after oocyte retrieval. Surplus embryos underwent blastocyst culture and vitrification. Moreover, luteal support was applied after ET.

1.3 Pregnancy Follow-up

Blood hCG tests were initially conducted 12–14 days after ET. Transvaginal ultrasound was performed 30–35 days after ET to confirm clinical pregnancy by the observation of a gestational sac and fetal heartbeat. Pregnancy and delivery complications and neonatal conditions were recorded until 6 weeks after birth.

1.4 Observation Indicators/Main Outcome Measures

Basic clinical data including age, body mass index (BMI), infertility duration, Gn dose, duration of stimulation, fertilization methods, endometrial thickness and basic sex hormone were collected. The primary outcome measure was clinical pregnancy. Pre-specified secondary outcomes included the process of ET, biochemical pregnancy rate, implantation rate, live birth rate, abortion rate, ectopic pregnancy rate, singleton live birth, pregnancy and neonatal complications.

1.5 Statistical Analysis

Variables were assessed for normality to determine whether parametric or nonparametric statistical methods should be used. Student's *t* test and the Mann-Whitney *U* test were used for comparing the central tendencies of parametric and nonparametric variables, respectively. The qualitative variables were compared by the χ^2 test or Fisher's exact test. The inter-group comparison of mean values was made using one-way ANOVA and between groups with SNK-*q* test. Statistical analysis was performed using SPSS 17.0 software package. All analyses of significance were two-sided and tested at the 5% level. $P < 0.05$ was considered statistically significant.

2 RESULTS

2.1 General Information

The general characteristics of the patients in both the groups are shown in table 1. No statistically significant differences were found in ovarian response-related factors such as baseline follicle-stimulating hormone (FSH), mean E₂ level on hCG day, duration of Gn stimulation, dosage of Gn and number of follicles with a diameter ≥ 14 mm on hCG day between the two groups. The presence of a previous CS scar did not impact ovar-

ian reserve function and ovarian response.

In previous CS group, 2 patients underwent CS twice, 9 cases presented with abnormal uterine bleeding/spotting and painful menstruation, 8 patients had post-cesarean section scar diverticulum (PCSD) which was confirmed by ultrasound imagining before or in the course of control ovarian hyperstimulation (COH), and

20 cases experienced intermittent or continuous presence of a little intrauterine fluid. Two patients underwent PCSD surgical treatment (one given hysteroscopic roller-ball coagulation of scar tissue and the other subject to combined laparoscopic and hysteroscopic repair of PCSD).

Table 1 The general clinical characteristics of patients in two groups

Parameters	CS group (n=144)	VD group (n=166)	P value
Age (years)	33.98±3.03	34.13±3.11	0.051
Infertility duration (years)	5.33±3.50	6.11±4.37	0.084
BMI (kg/m ²)	23.56±3.26	23.18±3.07	0.231
Basal FSH (IU/L)	7.19±2.52	7.22±2.33	0.901
Endometrial thickness (mm)	10.29±2.04	10.43±2.10	0.536
E ₂ level on hCG day (pg/mL)	3639.41±1886.02	3390.75±1703.29	0.224
Gn dosage (IU)	32.66±9.50	33.11±9.36	0.675
Duration of stimulation (days)	10.35±1.54	10.16±1.36	0.246
Numbers of follicles with a diameter ≥14 mm on hCG day	14.27±3.83	13.78±4.04	0.167

2.2 IVF/ICSI-ET Characteristics and Pregnancy Outcomes

For the comparison of embryo development-related factors between the two groups, the fertilization rate, cleavage rate, high-quality embryo rate, embryo cryopreservation cycle rate, the rate of multiple pregnancies, abortion rate and ectopic pregnancy rate were recorded and compared. But the pregnancy rates (40.28% vs. 54.22%) and the implantation rate (24.01% vs. 34.67%) in CS group were significantly lower than in VD group ($P>0.05$) (table 2).

In CS group, only 1 case had pregnancy after IVF-ET in 8 patients complicated with PCSD, while pregnancy was achieved in 8 cases in 20 patients detected with endometrial fluid by ultrasound. The preg-

nancy rates were significantly lower in CS+PCSD group and CS+endometrial fluid group than in VD group (12.5% vs. 40% vs. 54.82%, respectively).

In our study, we compared the difficult ET rate and catheter transfer depth between the two groups. The definition of difficult ET included: (1) longer time for transfer; (2) the use of a malleable adjustable stylet; (3) the application of a tenaculum; (4) the insertion of a hysterometer; (5) repeated ET. As demonstrated in table 3, there is statistically significant difference between patients with a history of CS versus those with a history of only VD in difficult ET ratio (9/144 vs. 0/166, $P=0.001$) and catheter transfer depth >7 cm ratio (23/144 vs. 1/166, $P<0.001$).

Table 2 The comparison of pregnancy outcomes after IVF/ICSI-ET

Parameters	CS group (n=144)	VD group (n=166)	P value
Retrieved oocytes (n)	11.52±4.96	10.47±5.28	0.073
ICSI (n)	45	39	0.125
Fertilization rate (%)	78.47 (1232/1570)	77.34 (1290/1668)	0.419
Cleavage rate (%)	97.40 (1200/1232)	97.29 (1255/1290)	0.812
High-quality embryo rate (%)	65.04 (642/987)	67.91 (675/994)	0.765
Embryo cryopreservation cycle rate (%)	75.69 (109/144)	62.05 (103/166)	0.010
Transfer embryos (n)	1.94±0.258	1.95±0.252	0.963
Embryo implantation rate (%)	24.01 (67/279)	34.98 (113/323)	0.003
Clinical pregnancy rate (%)	40.28 (58/144)	54.82 (91/166)	0.011
Single pregnancy	72.41 (42/58)	69.23 (63/91)	0.678
Twins pregnancy	27.59 (16/58)	30.77 (28/91)	0.678
Early abortion rate (%)	17.24 (10/58)	13.19 (12/91)	0.496
Ectopic pregnancy rate (%)	1.72 (1/58)	0	0.204

Table 3 The comparison of ET characteristics

Parameters	CS group (n=144)	VD group (n=166)	P value
Easy transfer (n)	135	166	
Difficult transfer (n)	9	0	0.001
Transfer depth >7 cm (n)	23	1	<0.001

2.3 Delivery Outcomes in Patients after IVF/ICSI-ET

In this study, we followed up 310 patients in two

groups until 6th week after birth except two were lost during follow-up (1 case in each group). One patient

with twins pregnancy in the previous CS group was subjected to an operation of selective embryo reduction, while 6 patients in the previous VD group underwent spontaneous reduction of twins to a singleton pregnancy.

There was no significant difference in live birth, rates of preterm birth, placenta previa, postpartum hemorrhage, birth weight, or gestational age of delivery in single birth between CS group and VD group ($P>0.05$).

The risk of pernicious placenta previa and postpartum hemorrhage in twin deliveries was significantly increased in CS group as compared with that in VD group ($P<0.05$), and gestational age and neonatal birth weight were significantly reduced in twin deliveries as compared with singleton deliveries in both the two groups ($P<0.05$) (fig. 1, and table 4).

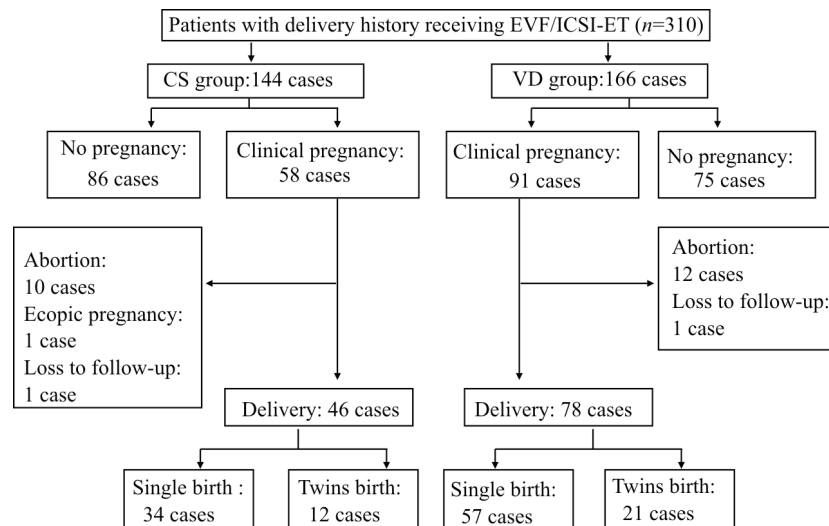


Fig. 1 Flow chart of the reproductive outcomes of IVF-ET in both two groups

Table 4 Delivery outcome and neonatal conditions after IVF-ET

Parameters	CS group		VD group	
	Single birth	Twins birth	Single birth	Twins birth
Cases (n)	34	12	57	21
Live birth (n)	34	24	57	42
Gestational age of delivery (weeks)	38.46±1.73	35.71±2.58*	38.95±1.46	35.83±2.24*
Full-term birth (n)	32	12	54	22
Preterm birth (n)	2	12*	3	20*
Stillbirth (n)	0	0	0	0
Birth weight (g)	3500±636.4	2618.75±453.2*	3385.6±462.8	2612.5±398.6*
Low birth weight (n)	3	9*	1	21*
Birth defects (n)	0	0	0	0
Placenta previa (n)	1	3 [#]	1	0
Postpartum hemorrhage (n)	0	4 [#]	0	1
Uterine rupture (n)	0	0	0	0

* $P<0.01$ vs. single birth groups, [#] $P<0.05$ vs. twins birth in VD group

3 DISCUSSION

With the implementation of full opening two-child policy in China, there has been an increase in the acceptance of IVF/ICIS-ET treatment among patients with previous cesarean delivery with secondary infertility. However, we are aware that it is more complex and dangerous for the future pregnancies in patients with uterine scar than intact uterus^[13], and so far, there is no clear evidence available whether the existence of CS scar negatively impacts the IVF-ET treatment and reproductive outcome. In our study, patients with a uterine scar had similar baseline FSH, mean E_2 level on hCG day, endometrial thickness, duration of Gn stimulation, and dosage of Gn, retrieved oocytes rate, fertilization rate, cleavage rate, high-quality embryo rate than VD

group, which suggests that the existence of a CS scar does not affect ovarian reserve function, ovarian response and embryonic development. However, the pregnancy rate (40.28% vs. 54.22%) and implantation rate (24.01% vs. 34.67%) in CS group were significantly lower than in VD group, which suggests that the presence of a previous CS scar may impact embryo implantation and decrease clinical pregnancy rate.

Moreover, our observation found that pregnancy rate decreased in women with PCSD and endometrial fluid in CS group as compared with VD group (12.5% vs. 40% vs. 54.82%, respectively). The possible reasons for this decrease are accumulation of mucus or blood in the hysterotomy scar defect, leading to the presence of intrauterine fluid, which could impact embryo implantation and thus, reduce pregnancy outcomes as found in

patients with endometrial fluid with tubal factor infertility. In two reviews of the literature reported by Naji *et al*^[5, 14], it was concluded that CS scar affects the site of implantation and subsequent placental migration in future pregnancies. Another mechanism may be the impairment of junctional zone contractility secondary to oxidative stress at the site of scarred endometrium. Bulletti and Brosens showed that the existence of cesarean scar may disturb myometrial contractility, reduce uterine compliance and modify uterine vascularization^[15, 16].

PCSD is a common complication which is mainly caused by poor wound healing after CS. In most cases, the presence of PCSD has no clinical significance, but in some patients, it may cause abnormal uterine bleeding, dysmenorrhea, dyspareunia, pelvic pain or may be associated with secondary infertility^[17, 18]. In our study, 9 patients with previous CS had abnormal uterine bleeding, 8 patients were diagnosed with PCSD and 20 cases were detected with the intermittent or continuous presence of intrauterine fluid using ultrasound. Transvaginal ultrasonographic examination is an important tool for making early diagnosis possible, even in asymptomatic cases^[19]. Abnormal uterine spotting or intrauterine fluid are probably caused by impaired contractility of muscle fibers surrounding the niche^[20]. For patients with bothering symptoms caused by uterine scar defect, surgical intervention can be suggested before IVF.

The ratio of embryo difficult transfer (9/144 vs. 0/166) and transfer depth >7 cm (23/144 vs. 1/166) in previous CS group were increased as compared with those in previous VD group. Similarly, in 2016, Patounakis *et al*^[21] reported that ETs performed on patients with a history of CD are more difficult and require a significantly longer time. Some possible reasons related to ET with CS scar patients are as follows: (1) The old uterine scar forms a tent-like structure anterior to the lower uterine segment, the advancing ET catheter may repeatedly lodge in the uterine scar or diverticulum and miss the uterine cavity, especially in the retroverted uterus. (2) The lifting of uterus and cervix caused by tissue adhesions post-CS will increase the difficulties of cervix exposure and ET. In addition, elongation of the uterus causes deeper transfer location. (3) Changes in uterine flexion after cesarean delivery, especially in the presence of a deficient CS scar. Recently, a study by Ryo *et al*^[22] showed that the changes in uterine flexion were more frequent among women who had a cesarean delivery than those who had a vaginal delivery, especially from ante-flexion to post-flexion.

Currently, in China, the maximum number of embryos permitted to be transferred for the first IVF cycle is two, but with double-embryo IVF-ET, the risk of iatrogenic multiple births significantly increases. The negative effect of multiple pregnancy on the human reproductive process is no longer a matter for debate. Twin pregnancy in previous cesarean scar patients was associated with increased risk of placenta previa and uterine scar rupture, preterm birth, and reduced birth weights of twin infants^[23, 24]. In our study, twin deliveries had significantly lower gestational age and neonatal birth weight than singleton deliveries in both groups. Moreover, the risk of twin pregnancy complication with pernicious placenta previa and postpartum hemorrhage in previous CS group was significantly higher than VD

group.

In light of our findings, reducing the number of embryos transferred to CS patients is the most effective method to prevent iatrogenic multiple pregnancies and reduce maternal-fetal risk. More strenuous efforts must be continued towards reducing the incidence of multiple pregnancies, particularly in younger CS women seeking IVF. Physicians should guide patients properly for accepting high-quality single-ET or selective multiple embryo reduction to ensure a safer labor and delivery, as well as better neonatal outcome. Selective single-blastocyst transfer is another feasible option to prevent multiple pregnancies while maintaining the live-birth rate.

In view of the increasing rate of delivery by CS and particularity of multiple ET in assisted-reproduction techniques, appropriate counseling to women desiring fertility with a history of CS is essential. Physicians should maintain awareness of the possible complications of post-cesarean delivery and perform related examination before managing ART. TVS is a noninvasive and effective tool for use in evaluating the healing of the CS incision, measuring the size, shape, and position of PCSD, and detecting endometrial fluid^[25, 26]. When a large scar defect is found in a symptomatic woman (pelvic pain, bleeding uterine, infertility), a surgical repair can be proposed, especially for a woman who desires a new pregnancy.

In addition, contraception is strictly necessary for women after cesarean delivery. Customarily, the inter-pregnancy interval of 24 to 36 months is the best period for caesarean incision healing and low risk for uterine rupture^[27, 28]. Multiple prospective and retrospective studies^[29–32] had shown that patients with short inter-pregnancy interval, greater number of repeated CSs, multiple pregnancy and fetal macrosomia increase the patient's risk for morbidity mainly due to pathological placentation, uterine rupture, massive bleeding and other complications. A caesarean is known to present one of the risk factors for placental pathologies (i.e. placenta previa and placenta percreta) in the subsequent pregnancies^[33]. Our data also demonstrated that the number of placenta previa and postpartum hemorrhage in CS group of twins pregnancy is greater than in previous VD group.

In conclusion, this study demonstrated that the existence of CS scar may impact embryo implantation and clinical pregnancy outcome, and increase the difficulty of ET. We should pay attention to the examination and the assessment of cesarean scar healing before IVF and strengthen gestational supervision. Limiting the number of transfer embryo is necessary for patients with previous CS because of multiple pregnancies risk and scarred uterus gestational complications. Further studies on a larger scale will certainly be needed to confirm our findings.

Conflict of Interest Statement

The authors have no conflicts of interest to disclose.

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